

WESAM ALDIAB

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EDUCATION

Adelaide University

Adelaide, SA

Bachelor of Software Engineering (Honours)

Jul 2024 — Oct 2028

- Relevant coursework: Object-Oriented Programming, Software Engineering, Algorithm Design and Data Structures, Computer Systems, Structured Data, Systems Engineering, and Network Foundations.

PROJECTS

Nonogram Solver and Automation System (C++)

Jan 2026 – Present

- Developed a constraint-based Nonogram solver using AC-3 propagation, depth-first search, and backtracking to process puzzle constraints and generate validated solutions.
- Designed modular data structures, file-input handling, and testing workflows while diagnosing build, file-path, and logical errors.
- Currently extending the project to read puzzles directly from the screen, reconstruct the grid, solve it, and automatically play back the solution.

Nand2Tetris: 16-bit Computer and Software Stack

Jul 2025 – Oct 2025

- Built a functioning 16-bit computer from NAND gates, including logic gates, adders, registers, memory, and an arithmetic logic unit.
- Implemented software translation components including an assembler, virtual machine tools, and compiler-related stages, strengthening understanding of computer architecture and abstraction.

Alien Translations Educational Web Application

2025

- Developed an educational word-unscrambling game within a two-week Ruby on Rails group project supported by an SQL relational database.
- Implemented random word selection, configurable difficulty, interface feedback, game logic, and integration with the application's MVC structure and routes.
- Learned Ruby, Ruby on Rails, SQL, HTML, and CSS during development and used documentation, testing, and debugging output to resolve issues under a strict deadline.

Pong Game Physics Engine (C++)

Jul 2025 – Oct 2025

- Developed gameplay and physics components including movement, collision detection, and configurable behaviour.
- Used JSON configuration and hot reloading to adjust game settings without rebuilding the application.

EXTRACURRICULAR ACTIVITIES

Ravi's Study Program

Dec 2025 – Feb 2026

- Participated in an intensive program led by industry mentors, focusing on advanced data structures, algorithms, object-oriented design, and collaborative problem-solving.

Competitive Programming — ACPC × RSP

Feb 2026

- Worked with other students to solve timed programming challenges using algorithms, data structures, logical analysis, and efficient implementation techniques.

AWARDS

Principal Scholarship

Jan 2021

- Awarded in recognition of high academic achievement and significant contributions to the school and wider community.

SKILLS

Programming Languages: C++, JavaScript, Ruby, MATLAB, HTML, CSS, SQL

Frameworks and Tools: Ruby on Rails, Git, Excel, HDL, JSON

Software Engineering: Object-Oriented Programming, SOLID principles, design patterns, data structures and algorithms, relational databases, software testing, debugging, requirements analysis, and technical documentation

Modelling: UML class diagrams, sequence diagrams, and use-case diagrams

Languages: English and Arabic